LASSO for the force constants with exact invariances (FOCEX, `enforce_inv` = 2 and 3), and the error bars of the fitted FCs

FOCEX notes

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Abstract

FOCEX fits the force constants (FCs) to force–displacement data while imposing the translational, rotational and Huang invariances exactly. With `enforce_inv` = 2 and 3 the fit is a LASSO: an L_1 penalty on the FCs themselves selects the terms the data support; with 3 the selected FCs are then refitted by ordinary (filtered) least squares. This note derives each step from the starting problem, says which routine does it (file:routine), and ends with how the standard errors of the FCs are computed for every value of `enforce_inv`.

Notation

symbol	size	meaning
$\mathbf{x}$	n_x	irreducible FCs of all fitted ranks (the unknowns)
$\mathbf{A},\ \mathbf{b}$	$N \times n_x,\ N$	force equations: $\mathbf{Ax} \approx \mathbf{b}$ (N = number of force components)
$\mathbf{C},\ \mathbf{d}$	$n_C \times n_x$	invariance rows: $\mathbf{Cx} = \mathbf{d}$ ($\mathbf{d} \neq 0$ only for a stressed reference, Coulomb)
$\mathbf{K}$	$n_x \times n_K$	orthonormal basis of the kernel of $\mathbf{C}$ ($\mathbf{CK} = 0$, $\mathbf{K}^T\mathbf{K} = \mathbf{1}$)
$\mathbf{x}_0$	n_x	particular solution, $\mathbf{Cx}_0 = \mathbf{d}$ (0 if $\mathbf{d} = 0$)
$\sigma_k,\ \mathbf{u}_k,\ \mathbf{v}_k$		singular values / vectors of $\mathbf{AK}$
$\mathbf{K}_r$	$n_x \times r$	$\mathbf{KV}_r$: kernel basis rotated onto the right singular vectors of $\mathbf{AK}$
$w_j,\ \mathbf{W}$		rms of column j of $\mathbf{A}$; $\mathbf{W} = \mathrm{diag}(w_j)$
$\mathbf{z}$	n_x	$\mathbf{Wx}$, standardised FCs
$\mathbf{Q}$	$n_x \times r$	orthonormal basis of the range of $\mathbf{WK}_r$
$\mathbf{t}$	r	coordinates in that basis: $\mathbf{z} = \mathbf{z}_0 + \mathbf{Qt}$
S		the selected FCs (support of the LASSO solution)

1 The problem and the exact invariances

The data give N force equations and the invariances n_C linear relations,

$$\mathbf{Ax} \approx \mathbf{b}, \qquad \mathbf{Cx} = \mathbf{d}. \tag{1}$$

$\mathbf{C}$ is rank deficient with many redundant rows (exact cubic test model: $n_C = 1882$, rank 379, $n_x = 555$). `svd_ridge.f90:kernel_and_particular` takes the SVD $\mathbf{C} = \mathbf{U}_C\mathbf{\Sigma}_C\mathbf{V}_C^T$; the columns of $\mathbf{V}_C$ whose singular value is below `svdcut` $\times$ the largest form $\mathbf{K}$, and $\mathbf{x}_0 = \sum_{\text{others}} \mathbf{v}_{C,j}(\mathbf{u}_{C,j} \cdot \mathbf{d})/\sigma_{C,j}$ is the minimum-norm solution of $\mathbf{Cx} = \mathbf{d}$. Every vector of the form

$$\mathbf{x} = \mathbf{x}_0 + \mathbf{Ky} \tag{2}$$

satisfies the invariances, since $\mathbf{Cx} = \mathbf{Cx}_0 + (\mathbf{CK})\mathbf{y} = \mathbf{d} + 0$; conversely every solution of $\mathbf{Cx} = \mathbf{d}$ is of this form. So the constraint is removed by fitting the n_K free coefficients $\mathbf{y}$ instead of $\mathbf{x}$. Substituting (2) in the force equations,

$$\mathbf{Ax} - \mathbf{b} = \mathbf{AKy} - \mathbf{b}', \qquad \mathbf{b}' = \mathbf{b} - \mathbf{Ax}_0. \tag{3}$$

2 Reference: the least-squares fit (`enforce_inv` = 1) and its filter

`constrained_fit.f90:fit_kernel_basis` minimises $\|\mathbf{AKy} - \mathbf{b}'\|^2$ by `svd_ridge.f90:svd_set`. With the thin SVD $\mathbf{AK} = \mathbf{U\Sigma V}^T = \sum_k \sigma_k \mathbf{u}_k \mathbf{v}_k^T$, write $\mathbf{y} = \sum_k y_k \mathbf{v}_k$ and $\beta_k = \mathbf{u}_k \cdot \mathbf{b}'$. Because the $\mathbf{u}_k$ are orthonormal,

$$\|\mathbf{AKy} - \mathbf{b}'\|^2 = \sum_k (\sigma_k y_k - \beta_k)^2 + \|\mathbf{b}'\|^2 - \sum_k \beta_k^2, \tag{4}$$

so each y_k is fitted independently: $y_k = \beta_k/\sigma_k$. This is unstable when σ_k is small (noise in β_k is divided by σ_k). `svbksb` uses instead

$$y_k = \phi_k \frac{\beta_k}{\sigma_k}, \qquad \phi_k = \frac{\sigma_k^3}{\sqrt{\sigma_k^6 + \sigma_c^6}}, \qquad \sigma_c = \text{cut} \times \sigma_{\max}, \tag{5}$$

so $\phi_k \to 1$ for $\sigma_k \gg \sigma_c$ and $y_k \to \sigma_k^2 \beta_k/\sigma_c^3 \to 0$ for $\sigma_k \ll \sigma_c$. The cut is `svdcut` = (`default.params` line 2)^2, 10^{-10} by default, possibly moved up into a gap of the singular values (`svd_ridge.f90:gap_cut`), and raised once more if the measured noise $\|\mathbf{Ax} - \mathbf{b}\|^2/\|\mathbf{b}\|^2$ exceeds 100 `svdcut` and the result passes the noise test of §8 (`svd_set`).

The filter is a quadratic penalty. Minimising $\frac{1}{2}(\sigma_k y_k - \beta_k)^2 + \frac{1}{2} g_k y_k^2$ gives $y_k = \sigma_k \beta_k/(\sigma_k^2 + g_k)$. This equals (5), $y_k = \sigma_k^2 \beta_k/\sqrt{\sigma_k^6 + \sigma_c^6}$, when $\sigma_k^2 + g_k = \sqrt{\sigma_k^6 + \sigma_c^6}/\sigma_k$, i.e.

$$g_k = \sigma_k^2 \left(\sqrt{1 + (\sigma_c/\sigma_k)^6} - 1 \right) \qquad \left(\approx \sigma_c^6/2\sigma_k^4 \text{ for } \sigma_k \gg \sigma_c, \;\; \approx \sigma_c^3/\sigma_k \text{ for } \sigma_k \ll \sigma_c \right). \tag{6}$$

The LASSO uses this form (§4): it reproduces the least-squares fit exactly when $\lambda = 0$.

3 The constrained LASSO

$$\min_{\mathbf{x}} \; \frac{1}{2N} \|\mathbf{Ax} - \mathbf{b}\|^2 + \lambda \sum_j w_j |x_j| \qquad \text{subject to} \qquad \mathbf{Cx} = \mathbf{d}. \tag{7}$$

The penalty is on the FCs x_j, not on the kernel coefficients y_k: each column of $\mathbf{K}$ mixes many FCs, so a sparse $\mathbf{y}$ is a dense $\mathbf{x}$ and vice versa (§11). The weight $w_j = \left(\sum_i A_{ij}^2/N \right)^{1/2}$ (`lasso_constrained.f90:classo_setup`) is the rms force that FC j produces per unit value, so $w_j|x_j|$ is in units of force for every rank, and the penalty does not depend on the units of each rank. Columns the data do not see are floored, $w_j \geq 10^{-8} \max w$ (§5). The code solves (7) in five steps (`classo_cv`, `classo_cv_res`).

4 Step 1: rotate the kernel onto the singular directions of the data

`lasso_constrained.f90:resolved_kernel` computes the SVD $\mathbf{AK} = \mathbf{U\Sigma V}^T$ of §2 and returns

$$\mathbf{K}_r = \mathbf{KV}_r, \qquad \mathbf{V}_r = \{\mathbf{v}_k : \; \sigma_k > 10^{-15} \sigma_{\max}\}, \tag{8}$$

i.e. only exactly-null directions (forces do not change at all) are dropped; r is their number. $\mathbf{K}_r$ is again an orthonormal basis of (the resolved part of) the kernel, because $\mathbf{K}_r^T \mathbf{K}_r = \mathbf{V}_r^T \mathbf{K}^T \mathbf{K} \mathbf{V}_r = \mathbf{1}$, so (2) becomes $\mathbf{x} = \mathbf{x}_0 + \mathbf{K}_r \mathbf{y}$, with $\mathbf{y}$ now the coordinates along the $\mathbf{v}_k$. In these coordinates (4) holds term by term, and the filter of §2 is the diagonal penalty $\frac{1}{2N} \sum_k g_k y_k^2$ with g_k from (6) (`classo_cv_res:filter_penalty`). Without it, FCs invisible in the data cost almost nothing and absorb arbitrary values through the invariance couplings (Ge: phonons of $10^8\,\text{cm}^{-1}$).

How V is obtained without squaring σ (H and R). Forming $\mathbf{K}^T\mathbf{A}^T\mathbf{A}\mathbf{K}$ would be cheap, but its eigenvalues are σ_k^2: a ratio $\sigma_k/\sigma_{\max} = 10^{-10}$ becomes 10^{-20}, below the double-precision resolution 2×10^{-16}. Instead the rows of $\mathbf{AK}$ are fed one at a time and absorbed into an upper-triangular $r\times r$ matrix $\mathbf{R}$ by Givens rotations (2×2 rotations that zero one element of the incoming row against the diagonal of $\mathbf{R}$). The product of all rotations is an orthogonal $\mathbf{H}$ ($N\times N$, never stored) with $\mathbf{AK} = \mathbf{H}\begin{pmatrix}\mathbf{R}\\0\end{pmatrix}$. If $\mathbf{R} = \mathbf{U}_R\mathbf{\Sigma}\mathbf{V}^T$ (`svdcmp`, cheap: $r\times r$), then $\mathbf{AK} = \left(\mathbf{H}\begin{smallmatrix}\mathbf{U}_R\\0\end{smallmatrix}\right)\mathbf{\Sigma}\mathbf{V}^T$, and the first factor has orthonormal columns; so $\mathbf{R}$ has the same singular values and right singular vectors $\mathbf{V}$ as $\mathbf{AK}$, without any squaring. Cost about Nn_K^2, memory n_K^2; $\mathbf{AK}$ is never stored.

5 Step 2: variables in which the problem has no constraint and a simple penalty

After step 1, (7) reads

$$\min_{\mathbf{y}}\ \frac{1}{2N}\|\mathbf{AK}_r\mathbf{y}-\mathbf{b}'\|^2 + \frac{1}{2N}\sum_k g_k y_k^2 + \lambda\|\mathbf{W}(\mathbf{x}_0+\mathbf{K}_r\mathbf{y})\|_1. \tag{9}$$

The constraint is gone, but the L_1 term is a norm of a linear map of $\mathbf{y}$. Three changes of variable make it simpler.

(a) Standardised FCs. $\mathbf{z} = \mathbf{W}\mathbf{x}$, so $\sum_j w_j|x_j| = \|\mathbf{z}\|_1$ and $\mathbf{z} = \mathbf{z}_0 + \mathbf{W}\mathbf{K}_r\mathbf{y}$, $\mathbf{z}_0 = \mathbf{W}\mathbf{x}_0$.

(b) An orthonormal basis of the reachable $\mathbf{z}$. The vectors $\mathbf{W}\mathbf{K}_r\mathbf{y}$ span the range of the $n_x\times r$ matrix $\mathbf{W}\mathbf{K}_r$. Take its thin SVD (`classo_setup`), $\mathbf{W}\mathbf{K}_r = \mathbf{Q}\,\mathbf{\Sigma}_W\mathbf{V}_W^T$, with $\mathbf{Q}$ ($n_x\times r$, $\mathbf{Q}^T\mathbf{Q}=\mathbf{1}$) its left singular vectors, and define

$$\mathbf{t} = \mathbf{B}\mathbf{y}, \qquad \mathbf{B} = \mathbf{Q}^T\mathbf{W}\mathbf{K}_r = \mathbf{\Sigma}_W\mathbf{V}_W^T \ (r\times r,\ \text{invertible}), \qquad \text{so} \qquad \mathbf{z} = \mathbf{z}_0+\mathbf{Q}\mathbf{t}, \quad \mathbf{x} = \mathbf{x}_0+\mathbf{W}^{-1}\mathbf{Q}\mathbf{t}. \tag{10}$$

$\mathbf{B}$ is invertible because $\mathbf{W}$ is diagonal positive and $\mathbf{K}_r$ has orthonormal columns, so $\mathbf{W}\mathbf{K}_r$ has full column rank (the code checks $\min\sigma_W > 10^{-12}\max\sigma_W$, `info`$=1$ otherwise). The penalty becomes $\lambda\|\mathbf{z}_0+\mathbf{Q}\mathbf{t}\|_1$, and the orthonormal $\mathbf{Q}$ keeps the Newton systems of step 3 well scaled (any basis of the range would do mathematically).

(c) The data term in $\mathbf{t}$. With (3) and (10), $\mathbf{A}\mathbf{x}-\mathbf{b} = \mathbf{A}\mathbf{W}^{-1}\mathbf{Q}\mathbf{t}-\mathbf{b}'$. Define $\tilde{\mathbf{A}} = \mathbf{A}\mathbf{W}^{-1}\mathbf{Q}/\sqrt{N}$ ($N\times r$) and expand the square:

$$\frac{1}{2N}\|\mathbf{A}\mathbf{x}-\mathbf{b}\|^2 = \tfrac{1}{2}\|\tilde{\mathbf{A}}\mathbf{t}-\mathbf{b}'/\sqrt{N}\|^2 = \tfrac{1}{2}\mathbf{t}^T\mathbf{G}\mathbf{t} - \mathbf{c}^T\mathbf{t} + \tfrac{1}{2N}\|\mathbf{b}'\|^2, \qquad \mathbf{G} = \tilde{\mathbf{A}}^T\tilde{\mathbf{A}}, \ \ \mathbf{c} = \tilde{\mathbf{A}}^T\mathbf{b}'/\sqrt{N}. \tag{11}$$

$\mathbf{G}$ and $\mathbf{c}$ are accumulated row by row, $\mathbf{G} = \sum_i \tilde{\mathbf{a}}_i\tilde{\mathbf{a}}_i^T$ with $\tilde{\mathbf{a}}_i = \mathbf{Q}^T(\mathbf{A}_{i,:}/\mathbf{w})^T/\sqrt{N}$, separately for each cross-validation fold (step 5). The filter penalty in $\mathbf{t}$: $\mathbf{y} = \mathbf{B}^{-1}\mathbf{t}$, so $\frac{1}{2N}\sum_k g_k y_k^2 = \frac{1}{2}\mathbf{t}^T\mathbf{P}\mathbf{t}$ with $\mathbf{P} = \mathbf{B}^{-T}\mathrm{diag}(g_k)\mathbf{B}^{-1}/N$. Collecting, (7) is exactly

$$\boxed{\min_{\mathbf{t}}\ \tfrac{1}{2}\mathbf{t}^T(\mathbf{G}+\mathbf{P})\mathbf{t} - \mathbf{c}^T\mathbf{t} + \lambda\|\mathbf{z}_0+\mathbf{Q}\mathbf{t}\|_1} \tag{12}$$

with no constraint: every $\mathbf{t}$ gives FCs that satisfy the invariances exactly.

The largest useful λ. Take $\mathbf{z}_0 = 0$. At $\mathbf{t} = 0$ the L_1 term is not differentiable; $\mathbf{t} = 0$ is the minimum if 0 is in the subgradient, $0 \in -\mathbf{c} + \lambda\mathbf{Q}^T\boldsymbol{\nu}$ with $|\nu_i| \le 1$. The choice $\boldsymbol{\nu} = \mathbf{Q}\mathbf{c}/\lambda$ satisfies $\mathbf{Q}^T\boldsymbol{\nu} = \mathbf{Q}^T\mathbf{Q}\mathbf{c}/\lambda = \mathbf{c}/\lambda$, and $|\nu_i| \le 1$ when $\lambda \ge \lambda_{\max} \equiv \|\mathbf{Q}\mathbf{c}\|_\infty$. So for $\lambda \ge \lambda_{\max}$ all FCs are zero (this is a sufficient condition; the grid of step 5 starts there). With the L2 option (§8) $\mathbf{c}$ is replaced by $\mathbf{c} - \mathbf{c}_{pz}$.

Weight floor. $\mathbf{x} = \mathbf{x}_0 + \mathbf{W}^{-1}\mathbf{Q}\mathbf{t}$ divides by w_j. For a column at 10^{-20} the round-off of $\mathbf{Q}$ (10^{-16}) became FC errors of order 10^4 (Ge: an FC2 of 153 where the SVD has 10^{-6}). Hence $w_j \ge 10^{-8}\max w$; genuine Ge columns go down to $3\times10^{-6}\max w$, so only round-off zeros are affected.

6 Step 3: solving (12) by an interior-point method

`lasso_constrained.f90:ipm_classo`. Write $\mathbf{z} = \mathbf{z}_0 + \mathbf{Q}\mathbf{t}$ and introduce $\mathbf{s} \geq |\mathbf{z}|$ componentwise, i.e. $\mathbf{a} = \mathbf{s} - \mathbf{z} \geq 0$, $\mathbf{b} = \mathbf{s} + \mathbf{z} \geq 0$. Since the objective increases with s_i, at the optimum $s_i = |z_i|$, and (12) is the smooth quadratic program

$$\min_{\mathbf{t},\mathbf{s}} \ \tfrac{1}{2}\mathbf{t}^T\mathbf{G}\mathbf{t} - \mathbf{c}^T\mathbf{t} + \lambda\mathbf{1}^T\mathbf{s}, \qquad \mathbf{a} = \mathbf{s} - \mathbf{z} \geq 0, \quad \mathbf{b} = \mathbf{s} + \mathbf{z} \geq 0 \tag{13}$$

($\mathbf{G}$ here stands for $\mathbf{G} + \mathbf{P}$). With multipliers $\boldsymbol{\alpha}, \boldsymbol{\beta} \geq 0$, the Lagrangian $\mathcal{L} = \tfrac{1}{2}\mathbf{t}^T\mathbf{G}\mathbf{t} - \mathbf{c}^T\mathbf{t} + \lambda\mathbf{1}^T\mathbf{s} - \boldsymbol{\alpha}^T\mathbf{a} - \boldsymbol{\beta}^T\mathbf{b}$ gives the optimality (KKT) conditions

$$\partial_{\mathbf{t}}: \ \mathbf{G}\mathbf{t} - \mathbf{c} + \mathbf{Q}^T(\boldsymbol{\alpha} - \boldsymbol{\beta}) = 0, \qquad \partial_{\mathbf{s}}: \ \boldsymbol{\alpha} + \boldsymbol{\beta} = \lambda, \qquad \alpha_i a_i = \beta_i b_i = 0. \tag{14}$$

The interior-point method replaces the last pair by $\alpha_i a_i = \beta_i b_i = \mu > 0$ and drives $\mu \to 0$, taking Newton steps on (14). Linearising, with $\delta\mathbf{a} = \delta\mathbf{s} - \mathbf{Q}\delta\mathbf{t}$, $\delta\mathbf{b} = \delta\mathbf{s} + \mathbf{Q}\delta\mathbf{t}$ and residuals $\mathbf{r}_t$, $r_s = \lambda - \alpha - \beta$, r_a, r_b:

$$\begin{aligned} \alpha_i\delta a_i + a_i\delta\alpha_i = r_{a,i} \ &\Rightarrow \ \delta\alpha_i = r_{a,i}/a_i - A_i\delta a_i, \qquad A_i = \alpha_i/a_i, \\ \beta_i\delta b_i + b_i\delta\beta_i = r_{b,i} \ &\Rightarrow \ \delta\beta_i = r_{b,i}/b_i - B_i\delta b_i, \qquad B_i = \beta_i/b_i. \end{aligned}$$

Inserting these in $\delta\alpha_i + \delta\beta_i = r_{s,i}$ gives $\delta s_i = (e_i + (A_i - B_i)q_i)/(A_i + B_i)$ with $q_i = (\mathbf{Q}\delta\mathbf{t})_i$, $e_i = r_{a,i}/a_i + r_{b,i}/b_i - r_{s,i}$; then $\delta\alpha_i - \delta\beta_i = f_i + \frac{(A_i+B_i)^2-(A_i-B_i)^2}{A_i+B_i}q_i = f_i + D_i q_i$ with $f_i = r_{a,i}/a_i - r_{b,i}/b_i - (A_i - B_i)e_i/(A_i + B_i)$ and

$$D_i = \frac{4A_iB_i}{A_i + B_i} > 0. \tag{15}$$

The first equation of (14) then becomes one $r \times r$ system,

$$(\mathbf{G} + \mathbf{Q}^T\mathbf{D}\mathbf{Q})\,\delta\mathbf{t} = -\mathbf{r}_t - \mathbf{Q}^T\mathbf{f}, \tag{16}$$

positive definite even where $\mathbf{G}$ is badly conditioned ($\mathbf{D} > 0$, $\mathbf{Q}$ full column rank), solved by Cholesky (`chol_fac`, `chol_sol`); $\delta\mathbf{s}, \delta\boldsymbol{\alpha}, \delta\boldsymbol{\beta}$ follow from the lines above. Each iteration does it twice (Mehrotra predictor–corrector): a predictor with $r_a = -\alpha a$, $r_b = -\beta b$, then a corrector with $r_a = \sigma\mu - \alpha a - \delta\alpha^{\text{pred}}\delta a^{\text{pred}}$ (same for b), $\sigma = (\mu_{\text{pred}}/\mu)^3$; the step is 0.99 of the largest one that keeps $\mathbf{a}, \mathbf{b}, \boldsymbol{\alpha}, \boldsymbol{\beta} > 0$. It stops when the residuals of (14) and the duality gap $\boldsymbol{\alpha}^T\mathbf{a} + \boldsymbol{\beta}^T\mathbf{b}$ are below 10^{-10} (relative), in at most 300 iterations (10–16 in practice). For $\lambda \geq \|\mathbf{Q}\mathbf{c}\|_\infty$ and $\mathbf{z}_0 = 0$ it returns $\mathbf{t} = 0$ directly.

The selected FCs. $S = \{j : |z_j| > 10^{-8}\max|z|\}$ (`classo_path`). Interior-point zeros are small, not exact; at small λ a few terms that vanish in the exact solution pass the test (against LARS, unconstrained: 57 kept for 46 at $\lambda = 10^{-4}\lambda_{\max}$; no true term lost). The test errs on the side of keeping a term: a lost true term makes the refit collapse (§11), an extra one is fitted to about zero by it.

7 Step 4 (`enforce_inv` = 3): the refit on the selected FCs

What it means. Step 3 does two things at once: it *selects* the FCs in S (the others are zero) and it *shrinks* the values of those in S. The refit keeps the selection and discards the values: it solves the least-squares problem again, without the L_1 term, with the FCs outside S fixed at zero and the invariances still exact.

Why the values are biased. In one orthonormal direction the problem is $\min_x \frac{1}{2}(x-\beta)^2 + \lambda|x|$, whose solution is $x = \text{sign}(\beta)\max(|\beta| - \lambda, 0)$: every kept coefficient is too small in magnitude by λ. The least-squares value is β. The refit removes this bias (relaxed lasso, Meinshausen 2007).

The equations (`lasso_constrained.f90:refit_support`). Drop the L_1 term from (12) and impose $z_j = (\mathbf{z}_0 + \mathbf{Q}\mathbf{t})_j = 0$ for $j \notin S$. With $\mathbf{B}_S = \mathbf{Q}_{S^c,:}$ (the m rows of $\mathbf{Q}$ off the support, $m = |S^c|$) the constraint is $\mathbf{B}_S\mathbf{t} = -\mathbf{z}_{0,S^c}$. Its solutions are

$$\mathbf{t} = \mathbf{t}_p + \mathbf{N}\mathbf{s}, \qquad \mathbf{B}_S\mathbf{N} = 0, \quad \mathbf{N}^T\mathbf{N} = \mathbf{1}, \tag{17}$$

with $\mathbf{t}_p$ one solution (0 if $\mathbf{z}_0 = 0$) and the columns of $\mathbf{N}$ a basis of the null space of $\mathbf{B}_S$ (p free parameters). Substituting in $\frac{1}{2}\mathbf{t}^T\mathbf{G}\mathbf{t} - \mathbf{c}^T\mathbf{t}$ ($\mathbf{G}$ again including $\mathbf{P}$) and dropping constants,

$$\tfrac{1}{2}\mathbf{s}^T(\mathbf{N}^T\mathbf{G}\mathbf{N})\mathbf{s} - \big(\mathbf{N}^T(\mathbf{c} - \mathbf{G}\mathbf{t}_p)\big)^T\mathbf{s} \;\Rightarrow\; (\mathbf{N}^T\mathbf{G}\mathbf{N})\,\mathbf{s} = \mathbf{N}^T(\mathbf{c} - \mathbf{G}\mathbf{t}_p), \tag{18}$$

solved by the pseudo-inverse with eigenvalues below 10^{-13} of the largest dropped. Because $\mathbf{P}$ is included, this is the filtered least squares of §2 restricted to S; with S = all FCs it is `svd_set`'s fit.

The null space. $\mathbf{N}$ comes from a Householder QR of $\mathbf{B}_S^T$ ($r \times m$) with column pivoting (`null_space_qr`): $\mathbf{B}_S^T\mathbf{\Pi} = \mathbf{H}_{\rm QR}\begin{pmatrix}\mathbf{R}_{11} & \mathbf{R}_{12}\\ 0 & 0\end{pmatrix}$, the rank ρ being where the largest remaining column norm drops below 10^{-9} of the largest. Then $\mathbf{B}_S = \mathbf{\Pi}\begin{pmatrix}\mathbf{R}_{11}^T & 0\\ \mathbf{R}_{12}^T & 0\end{pmatrix}\mathbf{H}_{\rm QR}^T$, so the columns $\rho+1,\ldots,r$ of the orthogonal $\mathbf{H}_{\rm QR}$ satisfy $\mathbf{B}_S\mathbf{h}_j = 0$: they are $\mathbf{N}$, $p = r - \rho$. (An SVD routine did not converge on some $\mathbf{B}_S$, whose singular values cluster at 0 and 1.)

When to refit. With accurate forces the refit gives unbiased FCs (exact on the exact model). With noisy forces or too few snapshots it fits the selected terms to the noise; then `enforce_inv`$=2$ (no refit) is the safer choice. *Limitation:* the refit works with the Gram matrix $\mathbf{N}^T\mathbf{G}\mathbf{N}$, so its cut 10^{-13} on eigenvalues corresponds to $\sigma/\sigma_{\max} \approx 3\times10^{-7}$; directions between this and `svdcut` are treated as unresolved here.

8 Step 5: choosing λ

Grid. 40 values (`common/params.f90`: `lasso_nlambda`), log-spaced from $\lambda_{\max}$ to $10^{-4}\lambda_{\max}$ (`lasso_epsratio`); the path is solved from large to small λ, and the refit is redone only when S changes.

Cross-validation (default, `default.params` line 6 absent or $\neq 1$). The force rows are split into 5 folds, interleaved (row i in fold $1+$ mod $(i-1,5)$), so that every fold samples every snapshot. Since $\mathbf{G} = \sum_f \mathbf{G}_f$ over the folds, the training data of fold f are $\mathbf{G} - \mathbf{G}_f$, $\mathbf{c} - \mathbf{c}_f$ (no new pass over the data); multiplying by $N/(N - n_f)$ restores the $1/(2N_{\rm train})$ normalisation of (7). The held-out error is the mean squared residual per row of fold f, $(\mathbf{t}^T\mathbf{G}_f\mathbf{t} - 2\mathbf{c}_f^T\mathbf{t} + \|\mathbf{b}'_f\|^2/N)\,N/n_f$, for the model that is returned (the refit for 3, the LASSO for 2). The λ of smallest mean error is chosen; within $10^{-13}\langle b^2\rangle$ of the minimum (round-off ties, exact data) the largest such λ. If the minimum is at the end of the grid, the grid is extended by two decades at the same density, down to $10^{-8}\lambda_{\max}$ (`classo_cv`). The final FCs are the fit on all rows at that λ.

Noise test (`default.params` line 6 = 1). First the full fit on all directions (S = all, the filtered least squares) gives the residual $\mathrm{RSS}_{\rm full}$, its effective number of parameters $p_{\rm full}$ = tr of the hat matrix, and the noise $s^2 = \mathrm{RSS}_{\rm full}/(N - p_{\rm full})$. If the forces are Gaussian with variance s^2, dropping Δp parameters that are truly zero raises the residual by $s^2\chi^2_{\Delta p}$, of mean Δp and variance $2\Delta p$. The chosen λ is the largest one whose refit on all rows passes

$$\mathrm{RSS}(\lambda) - \mathrm{RSS}_{\rm full} \le s^2\big(\Delta p + 2\sqrt{2\Delta p}\big) \tag{19}$$

(mean + 2 standard deviations). The same test decides whether `svd_set` and `classo_cv_res` may raise the filter cut when the measured noise is larger than line 2 assumed.

Optional L2 term (`default.params` line 7 = 1): a Gaussian prior. If the forces have Gaussian noise of variance s^2 and each FC of rank ρ a Gaussian prior of width τ_ρ, the negative log posterior is $\|\mathbf{Ax}-\mathbf{b}\|^2/(2s^2)+\sum_j x_j^2/(2\tau_{\rho(j)}^2)$; multiplied by s^2/N it is the data term of (7) plus $\frac{s^2}{N}\sum_j x_j^2/(2\tau_{\rho(j)}^2)$. The code adds this term (maximum a posteriori, MAP), with s^2 from the full fit above and τ_ρ = rms of the rank-ρ FCs of that fit. In $\mathbf{t}$ (with $x_j = z_j/w_j$, $\mathbf{z}=\mathbf{z}_0+\mathbf{Qt}$) it is $\frac{1}{2}\mathbf{t}^T\mathbf{Q}^T\mathbf{D}_2\mathbf{Qt}+(\mathbf{Q}^T\mathbf{D}_2\mathbf{z}_0)^T\mathbf{t}$ + const, $D_{2,j}=s^2/(Nw_j^2\tau_j^2)$, so $\mathbf{P}\to\mathbf{P}+\mathbf{Q}^T\mathbf{D}_2\mathbf{Q}$ and $\mathbf{c}\to\mathbf{c}-\mathbf{c}_{pz}$, $\mathbf{c}_{pz}=\mathbf{Q}^T\mathbf{D}_2\mathbf{z}_0$. It enters the LASSO, the refits and the training fits; the CV errors and the noise test use the force residuals alone. (The L_1 term itself is the MAP of a Laplace prior.)

9 The error bars of the fitted FCs

Derivation (least squares). In (5), $\mathbf{y}=\mathbf{VFU}^T\mathbf{b}'$ with $\mathbf{F}=\mathrm{diag}\big(\sigma_k^2/\sqrt{\sigma_k^6+\sigma_c^6}\big)$. If $\mathbf{b}=\mathbf{b}_{\text{true}}+\boldsymbol{\epsilon}$ with independent errors of variance s^2 on every row, $\mathrm{Cov}(\boldsymbol{\epsilon})=s^2\mathbf{1}$ and

$$\mathrm{Cov}(\mathbf{y})=\mathbf{VFU}^T\,s^2\mathbf{1}\,\mathbf{UFV}^T=s^2\,\mathbf{V}\,\mathrm{diag}\Big(\frac{\sigma_k^4}{\sigma_k^6+\sigma_c^6}\Big)\mathbf{V}^T,\qquad \mathrm{Cov}(\mathbf{x})=\mathbf{K}\,\mathrm{Cov}(\mathbf{y})\,\mathbf{K}^T \tag{20}$$

($\mathbf{x}_0$ is fixed, so it adds no variance). For $\sigma_k \gg \sigma_c$ this is the usual $s^2(\mathbf{K}^T\mathbf{A}^T\mathbf{AK})^{-1}$. The noise is estimated from the fit itself, $s^2=\|\mathbf{Ax}-\mathbf{b}\|^2/(m-n)$, m rows and n parameters. The standard error of FC j is $\sigma_j=\sqrt{\mathrm{Cov}(\mathbf{x})_{jj}}$ (`svd_set`, `svbksb`, `fit_kernel_basis`).

By `enforce_inv`.

- **0** (invariance rows stacked on the force rows, one `svd_set` of $[\mathbf{C};\mathbf{A}]$, `focex.f90`): the formula above with $\mathbf{K}=\mathbf{1}$; $m=n_C+N$ counts the invariance rows as rows with the same noise as the forces.
- **1** (`fit_kernel_basis`): the formula above on $\mathbf{AK}$; $m=N$, $n=n_K$.
- **2 and 3** (`constrained_fit.f90:lasso_support_errors`, called after the LASSO): least squares on the selected FCs with the invariances exact, i.e. the formula above with $\mathbf{K}$ replaced by $\mathbf{KN}$, $\mathbf{N}$ the null space of the rows of $\mathbf{K}$ off the support (same S test, `null_space_qr`), and s^2 from the residual of that least-squares fit ($n=p$). For 3 this is the standard error of the returned refit. For 2 it is the error the refit would have; the returned FCs are the LASSO values (the refit is computed but not applied; the log gives max |refit − LASSO|). FCs outside S have $\sigma_j=0$ (their rows of $\mathbf{KN}$ vanish).
- `itemp` = 1 (Boltzmann-weighted fit): none, $\sigma_j=0$.

The log lists every irreducible FC with its σ_j, and sd(ρ) = the plain mean of σ_j over the irreducible FCs of rank ρ (`ios.f90:write_independent_fcs`); for 2 and 3 the zeros of the dropped FCs are included in that mean.

What these error bars do not contain. They are the covariance of a linear least-squares fit with independent equal noise on all rows. They do not include the uncertainty of the selection S, the LASSO shrinkage (for 2), the L2 prior (the error fit has no penalty), the choice of the cut, or model error (missing FCs, anharmonic terms beyond the fitted ranks): when the misfit is model error, s^2 measures it but its rows are not independent. No Bayesian posterior is computed; the standard errors coincide with the posterior standard deviations of a Gaussian likelihood with a flat prior, for directions well above the cut.

10 Tests

These were run on 24 October 2026 with the version of that date (hard cut `svdcut` in step 1 instead of the filter penalty, CV tie tolerance 10^{-10}); they have not been rerun for this revision.

Exact model. Pair potential $A(r_c^2 - r^2)^2$, fcc lattice with two species, ranks 2–4 fitted; its FCs are known exactly (FC2/FC3/FC4: 11/20/43 non-zero of 99/80/376). Forces from FOCEX's own matrix.

	FC2 max error	FC3 max error	FC4 max error	non-zero FC2/3/4
penalty on $\mathbf{y}$ (LARS solver)	28.8	0.67	288	98/80/376
`enforce_inv` = 3, exact forces	9×10^{-10}	4×10^{-9}	8×10^{-7}	11/20/43
1 % random noise on the forces:				
`enforce_inv` = 1 (SVD)	0.095	1.13	206	98/80/376
`enforce_inv` = 2	0.012	1.35	9.2	54/61/350
`enforce_inv` = 3	0.017	0.58	4.8	11/32/136

(largest FCs: FC2 80, FC3 48, FC4 288.) With noise this data set does not determine FC4 well for any method; the LASSO reduces the error by a factor 20–40 compared with the SVD.

Ge (FOCEX example, DFT forces). SVD: fit 0.036 %, 171 of 173 FCs non-zero. `enforce_inv` = 3: fit 0.044 %, 107 non-zero; FC2 within 2.5×10^{-4} and FC3 within 0.12 (largest 28) of the SVD values.

Solver checks. Interior point vs. cvxpy/Clarabel at five values of λ: objective equal to 10^{-11}; refit vs. numpy least squares: 4×10^{-9}.

11 Why these choices

- *Penalty on* $\mathbf{y}$ (the previous `enforce_inv` = 2, still available as `FOCEX_LASSO=old`): the columns of $\mathbf{K}$ are dense mixtures of FCs, so sparse FCs are dense in $\mathbf{y}$; on the exact model 98/80/376 non-zero for 11/20/43 true, FC4 wrong.
- *Elimination basis* (row-reduce $\mathbf{C}$; the free FCs are a subset of $\mathbf{x}$, as in ALAMODE): the result depends on which FCs are eliminated (those carry no penalty); FC4 error 28 of 288. ALAMODE with LASSO keeps only the constraints within each rank (`alm/constraint.cpp`); that left FC4 errors up to 1.2×10^4.
- *Free LASSO, then a constrained refit*: if the first step misses one true FC, the constraint rows containing it force its partners to zero in the refit (reduced kernel of dimension 3, FC3 = FC4 = 0). Hence the selection itself is done with the constraints (step 3).
- *Solver*: ADMM had not converged after 200 000 iterations at 3 of 5 values of λ; FISTA needs the proximal map of $\lambda\|\mathbf{Qt}\|_1$, which has no closed form; LARS does not handle the constraint. The interior point needs 10–16 iterations and agrees with cvxpy/Clarabel to 10^{-11}.

12 Open points

- The refit's Gram-matrix cut (§7) is coarser than `svdcut`; a QR-based refit would remove the difference.

- The selection threshold 10^{-8} (step 3) is a numerical choice, tested only on the cases above.
- The error bars of §9 leave out the selection uncertainty; a bootstrap over snapshots would include it.
- The tests should be rerun with the filter-penalty version.

References. R. Tibshirani, J. R. Stat. Soc. B **58**, 267 (1996) (LASSO; Laplace-prior MAP; cited from memory, not checked); B. Efron, T. Hastie, I. Johnstone, R. Tibshirani, Ann. Statist. **32**, 407 (2004) (LARS); S. Boyd *et al.*, Found. Trends Mach. Learn. **3**, 1 (2011) (ADMM); N. Meinshausen, Comput. Stat. Data Anal. **52**, 374 (2007) (relaxed lasso; cited from memory, not checked); S. Mehrotra, SIAM J. Optim. **2**, 575 (1992) (predictor–corrector; cited from memory, not checked); T. Tadano, ALAMODE, `github.com/ttadano/alamode`.